

items

course

Python

SQL

Database

Machine

All

two

final

step

items

Python

Python

Python

SQL, d

SQL

Database

Q1. Hospital Patient Records.

Total transactions = 5

Minimum support = 50%.

Minimum confidence = 70%.

Item	Transaction	Support.
fever	P ₁ , P ₂ , P ₄ , P ₅	4/5 = 80%.
cough	P ₁ , P ₃ , P ₄	3/5 = 60%.
headache	P ₂ , P ₃ , P ₄	3/5 = 60%.
fatigue	P ₅	1/5 = 20%.

fatigue is below 50%, so it is eliminated,

• fever + cough

Appears in P₁ & P₄

support (fever, cough) = 2/5 = 40%.

40% < 50%. this pair is not frequent.

• fever & headache.

Appears in P₂ & P₄

support (fever, headache) = 2/5 = 40%.

40% < 50%, this pair is not frequent.

1. Fever to cough

2. Fever to headache

3. fever to headache.

4. headache to fever.

fever $\rightarrow$ cough

$$\text{confidence} = \frac{\text{support}(\text{fever, cough})}{\text{support}(\text{fever})} = \frac{2/5}{4/5} = 50\%$$

50% < 70% so its fails

fever $\rightarrow$ headache.

$$\text{confidence} = \frac{2/5}{4/5} = 50\%$$

No valid association rule satisfies all the constraints. fever constraint is satisfied by the candidate rules, but their support is only 40%, which is below the required 50%. & confidence is only 50% below the required status 70%.

2. ONLINE Learning platform:

Minimum support = 40%.

Number of students = 5.

Minimum frequency = 40% $\times$ 5 = 2

an itemset must occur at least 2 transactions.

items

course	students	support
Python	S1, S2, S4, S5	$4/5 = 80\%$
SQL	S1, S3, S4	$3/5 = 60\%$
Data science	S2, S4	$2/5 = 40\%$
Machine learning	S3, S5	$2/5 = 40\%$

All courses are frequent, the constraint requires at least two courses, so individual courses are not included in the final items

Step 2: 2-items

itemset	students	support
Python, SQL	S1, S4	40%
Python, Data science	S2, S4	40%
Python, ML	S5	20%
SQL, data science	S4	20%
SQL, ML	S3	20%
Data science, ML	None	0%

frequent 2-itemsets are:

$\{\text{python}, \text{sql}\} \rightarrow 40\%$

$\{\text{python}, \text{DataScience}\} \rightarrow 40\%$

Step 2; 3-itemsets.

The only possible combinations do not occur at least twice

There are no frequent 3-itemsets

$\left. \begin{array}{l} \{\text{python}, \text{sql}\} \\ \{\text{python}, \text{DataScience}\} \end{array} \right\}$ Both have 40% support.

→ Minimum itemset size = 2 remove all 1-itemsets

3. Library Borrowing Records.

Minimum support is not explicitly given

minimum support = 40%, i.e. an itemset must occur in at least 2 out of 5 transactions,

Maximum itemset size = 3

Step 1: 1-itemsets.

Item	Transactions	Support
Data Mining	T_1, T_2, T_4, T_5	80%
Python programming	T_1, T_3, T_4	60%
Databases systems	T_2, T_3, T_4	60%
A_1	T_5	20%

A_1 is removed because its support is below 40%.

Step 2: 2-itemsets

Data mining + python programming.

occurs in $T_2 \cup T_4$:

$$\text{support} = \frac{2}{5} = 40\%$$

occurs in $T_1 \cup T_4$:

$$\text{support} = \frac{2}{5} = 40\%$$

Python pgm + database system

occurs in $T_3 \cup T_4$.

$$\text{support} = \frac{2}{5} = 40\%$$

{data mining, python pgm}

{data mining, database system}

{python pgm, database system}

step 3: 3-itemsets

{data mining, python pgm, database systems}

It occurs only in T_u

$$\text{support} = 1/5 = 20\%$$

The frequent items are:

- Data mining - 80%
- python pgm - 60%
- Database systems - 60%

2-itemsets:

- {Data mining, python pgm} - 40%
- {Data Mining, Database system} - 40%
- {python pgm, Database system} - 40%

3-itemsets: None

Effect of maximum size constraint

$$\text{Maximum size} = 3$$

Q4 computer science occurs in

- S₁
- S₃
- S₄

$$\text{support}(\text{computer science}) = 3/5 = 60\%$$

ML occurs with CS in

Q4

Q4

$$\text{support}(CS, ML) = 2/5 = 40\%$$

$$\text{support}(CS \rightarrow ML) = 2/5$$

$$2/5 = 40\%$$

$$\text{confidence}(CS \rightarrow ML) = \frac{\text{support}(CS, ML)}{\text{support}(CS)}$$

$$= \frac{2/5}{3/5} = 2/3 = 66.67\%$$

$$CS \rightarrow ML$$

with 40% support & 66.67% confidence.

$$ML \rightarrow CS$$

has support = 40%.

$$\text{confidence} = \frac{2/5}{2/5} = 100\%$$

$$ML \rightarrow CS$$

40% support

100% confidence.

Q5 Insurance customer Records.

Ag group = Young

customer	Age	Income	Policy.
C1	Young	high	health insurance
C3	young	low	Travel insurance
C5	young	medium	vehicle insurance.

young customer = 3

step 1: Income pattern.

Young + high income support = $1/3 = 33.33\%$

Youngs + low income
support = $1/3 = 33.33\%$

Young + medium income
support = $1/3 = 33.33\%$

step 3: Multidimensional patterns:

Age group	income	policy
Young	high	health insurance
Young	low	Travel insurance
Young	medium	vehicle insurance

young & high income $\rightarrow$ health insurance

Young & low income $\rightarrow$ Travel insurance.

young & medium income $\rightarrow$ vehicle insurance ID

1 out of 3 young customers.

$$\text{support} = 33.33\%$$

$$\text{confidence} = 100\%$$

confidence (young, high income $\rightarrow$ health insurance)

$$= \frac{1/3}{1/3} = 100\%$$

young + high income $\rightarrow$ health insurance
+ promote premium health plans.

young + low income $\rightarrow$

offer affordable travel policies

young + medium income $\rightarrow$, promote vehicle insurance, packages